Mzansi Couriers Relay Launch Simulation

Generated from the production shared-pool allocator and live road-route geometry. Customer prices, fuel, toll, backhaul, and payment-processing values are explicit scenario assumptions; route distances are resolved through the configured Maps directions proxy.

> **Important accounting limitation:** “Mzansi estimated net contribution” means the 20% platform fee less modeled payment-processing costs only. It is not accounting profit: staff/support, insurance, tax, refunds/chargebacks, map/API charges, payout fees, marketing, fraud, vehicle recovery, and platform maintenance are excluded.

Executive simulation summary

| Scenario | Revenue | Mzansi 20% | Courier pool | Couriers | Road km / legs | Route cost | Platform net | Decision | | Sandton → Soweto · 1 parcel(s) | R180.00 | R36.00 | R144.00 | 1 | 41 / 1 | R277.30 | R29.28 | REJECT / CONSOLIDATE | | Sandton → Soweto · 5 parcel(s) | R750.00 | R150.00 | R600.00 | 1 | 41 / 1 | R277.30 | R126.75 | DISPATCH | | Sandton → Soweto · 10 parcel(s) | R1500.00 | R300.00 | R1200.00 | 1 | 41 / 1 | R277.30 | R255.00 | DISPATCH | | Sandton → Soweto · 20 parcel(s) | R3000.00 | R600.00 | R2400.00 | 1 | 41 / 1 | R277.30 | R511.50 | DISPATCH | | JHB → Bloemfontein · 1 parcel(s) | R650.00 | R130.00 | R520.00 | 2 | 510 / 2 | R3638.97 | R109.65 | REJECT / CONSOLIDATE | | JHB → Bloemfontein · 5 parcel(s) | R3250.00 | R650.00 | R2600.00 | 2 | 510 / 2 | R3638.97 | R554.25 | REJECT / CONSOLIDATE | | JHB → Bloemfontein · 10 parcel(s) | R4500.00 | R900.00 | R3600.00 | 2 | 510 / 2 | R3638.97 | R768.00 | REJECT / CONSOLIDATE | | JHB → Bloemfontein · 20 parcel(s) | R9000.00 | R1800.00 | R7200.00 | 2 | 510 / 2 | R3638.97 | R1537.50 | DISPATCH | | JHB → Bloemfontein · 50 parcel(s) | R22500.00 | R4500.00 | R18000.00 | 2 | 510 / 2 | R3638.97 | R3846.00 | REJECT / CONSOLIDATE | | JHB → Bloemfontein · 100 parcel(s) | R45000.00 | R9000.00 | R36000.00 | 2 | 510 / 2 | R3638.97 | R7693.50 | DISPATCH | | JHB → Cape Town · 5 legs · 1 parcel(s) | R2800.00 | R560.00 | R2240.00 | 5 | 1421 / 5 | R10051.75 | R477.30 | REJECT / CONSOLIDATE | | JHB → Cape Town · 5 legs · 5 parcel(s) | R9500.00 | R1900.00 | R7600.00 | 5 | 1421 / 5 | R10051.75 | R1623.00 | REJECT / CONSOLIDATE | | JHB → Cape Town · 5 legs · 10 parcel(s) | R19000.00 | R3800.00 | R15200.00 | 5 | 1421 / 5 | R10051.75 | R3247.50 | DISPATCH | | JHB → Cape Town · 5 legs · 20 parcel(s) | R31000.00 | R6200.00 | R24800.00 | 5 | 1421 / 5 | R10051.75 | R5299.50 | DISPATCH | | JHB → Cape Town · 5 legs · 50 parcel(s) | R65000.00 | R13000.00 | R52000.00 | 5 | 1421 / 5 | R10051.75 | R11113.50 | REJECT / CONSOLIDATE | | JHB → Cape Town · 5 legs · 100 parcel(s) | R130000.00 | R26000.00 | R104000.00 | 5 | 1421 / 5 | R10051.75 | R22228.50 | DISPATCH | | JHB → Cape Town · 5 legs · 120 parcel(s) | R156000.00 | R31200.00 | R124800.00 | 5 | 1421 / 5 | R10051.75 | R26674.50 | DISPATCH | | JHB → Cape Town · uneven 4-leg split · 20 parcels | R33000.00 | R6600.00 |

R26400.00 | 4 | 1420 / 4 | R9990.04 | R5641.50 | DISPATCH | | JHB → Cape Town · 6 legs · 20 parcels | R36000.00 | R7200.00 | R28800.00 | 6 | 1519 / 6 | R10761.78 | R6154.50 | DISPATCH | | JHB → Cape Town · high fuel + empty returns | R33000.00 | R6600.00 | R26400.00 | 5 | 1421 / 5 | R19392.91 | R5641.50 | DISPATCH | | JHB → Cape Town · strong backhaul | R31000.00 | R6200.00 | R24800.00 | 5 | 1421 / 5 | R10801.75 | R5299.50 | DISPATCH | | JHB → Cape Town · one courier cancels / recovery return | R33000.00 | R6600.00 | R26400.00 | 5 | 1421 / 5 | R15417.75 | R5641.50 | DISPATCH | | JHB → Cape Town · 50% refund / failed delivery | R16500.00 | R3300.00 | R13200.00 | 5 | 1421 / 5 | R10051.75 | R2820.00 | REJECT / CONSOLIDATE |

Sandton → Soweto · 1 parcel(s)

- Customer revenue: **R180.00**; Mzansi gross platform revenue: **R36.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R29.28**.
- Total courier pool: **R144.00**; route cost floor: **R277.30**; route surplus after courier cost floor: **R-133.30**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **fails**; dispatch-time gate: **REPRICE**; decision: **REJECT / CONSOLIDATE**.
- Minimum customer tariff at this parcel count: **R346.63 per parcel**; minimum courier pool: **R277.30**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R346.63 per parcel** (courier cost floor R346.63, positive platform net R8.77).
- Readiness reasons: Courier pool does not cover operating cost plus disruption reserve At least one courier is below the R150.00 net acceptance floor; warnings: Courier pool is R133.30 below the combined operating-cost floor after deadhead, return, and handover costs.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Soweto | 40.5 | 1 | 2 | 40.5 | R0.00 | 10 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | local-1-leg-1 | 40.5 | R144.00 | R277.30 | R-133.30 | R100.00 | NO |

Sandton → Soweto · 5 parcel(s)

- Customer revenue: **R750.00**; Mzansi gross platform revenue: **R150.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R126.75**.
- Total courier pool: **R600.00**; route cost floor: **R277.30**; route surplus after courier cost floor: **R322.70**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R69.33 per parcel**; minimum courier pool: **R277.30**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R69.33 per parcel** (courier cost floor R69.33, positive platform net R1.75).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Soweto | 40.5 | 5 | 10 | 40.5 | R0.00 | 10 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | local-5-leg-1 | 40.5 | R600.00 | R277.30 | R322.70 | R100.00 | YES |

Sandton → Soweto · 10 parcel(s)

- Customer revenue: **R1500.00**; Mzansi gross platform revenue: **R300.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R255.00**.
- Total courier pool: **R1200.00**; route cost floor: **R277.30**; route surplus after courier cost floor: **R922.70**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R34.67 per parcel**; minimum courier pool: **R277.30**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R34.66 per parcel** (courier cost floor R34.66, positive platform net R0.88).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Soweto | 40.5 | 10 | 20 | 40.5 | R0.00 | 10 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | local-10-leg-1 | 40.5 | R1200.00 | R277.30 | R922.70 | R100.00 | YES |

Sandton → Soweto · 20 parcel(s)

- Customer revenue: **R3000.00**; Mzansi gross platform revenue: **R600.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R511.50**.
- Total courier pool: **R2400.00**; route cost floor: **R277.30**; route surplus after courier cost floor: **R2122.70**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R17.34 per parcel**; minimum courier pool: **R277.30**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R17.33 per parcel** (courier cost floor R17.33, positive platform net R0.44).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Soweto | 40.5 | 20 | 40 | 40.5 | R0.00 | 10 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | local-20-leg-1 | 40.5 | R2400.00 | R277.30 | R2122.70 | R100.00 | YES |

JHB → Bloemfontein · 1 parcel(s)

- Customer revenue: **R650.00**; Mzansi gross platform revenue: **R130.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R109.65**.

- Total courier pool: **R520.00**; route cost floor: **R3638.97**; route surplus after courier cost floor: **R-3118.97**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **fails**; dispatch-time gate: **REPRICE**; decision: **REJECT / CONSOLIDATE**.
- Minimum customer tariff at this parcel count: **R4548.72 per parcel**; minimum courier pool: **R3638.97**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R4548.71 per parcel** (courier cost floor R4548.71, positive platform net R8.77).
- Readiness reasons: Courier pool does not cover operating cost plus disruption reserve At least one courier is below the R150.00 net acceptance floor; warnings: Courier pool is R3118.97 below the combined operating-cost floor after deadhead, return, and handover costs.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Pretoria | 55.8 | 1 | 3 | 55.8 | R35.00 | 15 | none | 20 parcels / 150kg | | Pretoria → Bloemfontein | 454.6 | 1 | 3 | 454.6 | R160.00 | 25 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | regional-1-leg-1 | 55.8 | R136.38 | R418.35 | R-281.97 | R100.00 | NO | | courier-2 | regional-1-leg-2 | 454.6 | R383.62 | R3220.62 | R-2837.00 | R100.00 | NO |

JHB → Bloemfontein · 5 parcel(s)

- Customer revenue: **R3250.00**; Mzansi gross platform revenue: **R650.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R554.25**.
- Total courier pool: **R2600.00**; route cost floor: **R3638.97**; route surplus after courier cost floor: **R-1038.97**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **fails**; dispatch-time gate: **REPRICE**; decision: **REJECT / CONSOLIDATE**.
- Minimum customer tariff at this parcel count: **R909.75 per parcel**; minimum courier pool: **R3638.97**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R909.74 per parcel** (courier cost floor R909.74, positive platform net R1.75).

- Readiness reasons: Courier pool does not cover operating cost plus disruption reserve At least one courier is below the R150.00 net acceptance floor; warnings: Courier pool is R1038.97 below the combined operating-cost floor after deadhead, return, and handover costs.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Pretoria | 55.8 | 5 | 15 | 55.8 | R35.00 | 15 | none | 20 parcels / 150kg | | Pretoria → Bloemfontein | 454.6 | 5 | 15 | 454.6 | R160.00 | 25 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | regional-5-leg-1 | 55.8 | R372.86 | R418.35 | R-45.49 | R100.00 | NO | | courier-2 | regional-5-leg-2 | 454.6 | R2227.14 | R3220.62 | R-993.48 | R100.00 | NO |

JHB → Bloemfontein · 10 parcel(s)

- Customer revenue: **R4500.00**; Mzansi gross platform revenue: **R900.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R768.00**.
- Total courier pool: **R3600.00**; route cost floor: **R3638.97**; route surplus after courier cost floor: **R-38.97**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **fails**; dispatch-time gate: **REPRICE**; decision: **REJECT / CONSOLIDATE**.
- Minimum customer tariff at this parcel count: **R454.88 per parcel**; minimum courier pool: **R3638.97**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R454.87 per parcel** (courier cost floor R454.87, positive platform net R0.88).
- Readiness reasons: Courier pool does not cover operating cost plus disruption reserve At least one courier is below the R150.00 net acceptance floor; warnings: Courier pool is R38.97 below the combined operating-cost floor after deadhead, return, and handover costs.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Pretoria | 55.8 | 10 | 30 | 55.8 | R35.00 | 15 | none | 20 parcels / 150kg | | Pretoria → Bloemfontein | 454.6 | 10 | 30 | 454.6 | R160.00 | 25 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | regional-10-leg-1 | 55.8 | R486.55 | R418.35 | R68.20 | R100.00 | YES | | courier-2 | regional-10-leg-2 | 454.6 | R3113.45 | R3220.62 | R-107.17 | R100.00 | NO |

JHB → Bloemfontein · 20 parcel(s)

- Customer revenue: **R9000.00**; Mzansi gross platform revenue: **R1800.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R1537.50**.
- Total courier pool: **R7200.00**; route cost floor: **R3638.97**; route surplus after courier cost floor: **R3561.03**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R227.44 per parcel**; minimum courier pool: **R3638.97**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R227.44 per parcel** (courier cost floor R227.44, positive platform net R0.44).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Pretoria | 55.8 | 20 | 60 | 55.8 | R35.00 | 15 | none | 20 parcels / 150kg | | Pretoria → Bloemfontein | 454.6 | 20 | 60 | 454.6 | R160.00 | 25 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | regional-20-leg-1 | 55.8 | R895.84 | R418.35 | R477.49 | R100.00 | YES | | courier-2 | regional-20-leg-2 | 454.6 | R6304.16 | R3220.62 | R3083.54 | R100.00 | YES |

JHB → Bloemfontein · 50 parcel(s)

- Customer revenue: **R22500.00**; Mzansi gross platform revenue: **R4500.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R3846.00**.
- Total courier pool: **R18000.00**; route cost floor: **R3638.97**; route surplus after courier cost floor: **R14361.03**.

- Capacity feasible: **no**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **CONSOLIDATE**; decision: **REJECT / CONSOLIDATE**.
- Parallel planning: **3 wave(s)**; wave decisions: W1 20 parcels / R7200.00 pool / R3561.03 courier net, W2 20 parcels / R7200.00 pool / R3561.03 courier net, W3 10 parcels / R3600.00 pool / R-38.97 courier net; planner result: **CONSOLIDATE**.
- Minimum customer tariff at this parcel count: **R90.98 per parcel**; minimum courier pool: **R3638.97**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R90.97 per parcel** (courier cost floor R90.97, positive platform net R0.18).
- Readiness reasons: Vehicle capacity, parcel count, weight, or volume is not feasible; warnings: One or more relay legs exceed the vehicle's verified parcel, weight, or volume capacity.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Pretoria | 55.8 | 50 | 150 | 55.8 | R35.00 | 15 | none | 20 parcels / 150kg | | Pretoria → Bloemfontein | 454.6 | 50 | 150 | 454.6 | R160.00 | 25 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | regional-50-leg-1 | 55.8 | R2123.70 | R418.35 | R1705.35 | R100.00 | YES | | courier-2 | regional-50-leg-2 | 454.6 | R15876.30 | R3220.62 | R12655.68 | R100.00 | YES |

JHB → Bloemfontein · 100 parcel(s)

- Customer revenue: **R45000.00**; Mzansi gross platform revenue: **R9000.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R7693.50**.
- Total courier pool: **R36000.00**; route cost floor: **R3638.97**; route surplus after courier cost floor: **R32361.03**.
- Capacity feasible: **no**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **CONSOLIDATE**; decision: **DISPATCH**.
- Parallel planning: **5 wave(s)**; wave decisions: W1 20 parcels / R7200.00 pool / R3561.03 courier net, W2 20 parcels / R7200.00 pool / R3561.03 courier net, W3 20 parcels / R7200.00 pool / R3561.03 courier net, W4 20 parcels / R7200.00 pool / R3561.03 courier net, W5 20 parcels / R7200.00 pool / R3561.03 courier net; planner result: **DISPATCH**.

- Minimum customer tariff at this parcel count: **R45.49 per parcel**; minimum courier pool: **R3638.97**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R45.49 per parcel** (courier cost floor R45.49, positive platform net R0.09).
- Readiness reasons: Vehicle capacity, parcel count, weight, or volume is not feasible; warnings: One or more relay legs exceed the vehicle’s verified parcel, weight, or volume capacity.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Pretoria | 55.8 | 100 | 300 | 55.8 | R35.00 | 15 | none | 20 parcels / 150kg | | Pretoria → Bloemfontein | 454.6 | 100 | 300 | 454.6 | R160.00 | 25 | none | 20 parcels / 150kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | regional-100-leg-1 | 55.8 | R4170.14 | R418.35 | R3751.79 | R100.00 | YES | | courier-2 | regional-100-leg-2 | 454.6 | R31829.86 | R3220.62 | R28609.24 | R100.00 | YES |

JHB → Cape Town · 5 legs · 1 parcel(s)

- Customer revenue: **R2800.00**; Mzansi gross platform revenue: **R560.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R477.30**.
- Total courier pool: **R2240.00**; route cost floor: **R10051.75**; route surplus after courier cost floor: **R-7811.75**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **fails**; dispatch-time gate: **REPRICE**; decision: **REJECT / CONSOLIDATE**.
- Minimum customer tariff at this parcel count: **R12564.69 per parcel**; minimum courier pool: **R10051.75**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R12564.69 per parcel** (courier cost floor R12564.69, positive platform net R8.77).
- Readiness reasons: Courier pool does not cover operating cost plus disruption reserve At least one courier is below the R150.00 net acceptance floor; warnings: Courier pool is R7811.75 below the combined operating-cost floor after deadhead, return, and handover costs.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 1 | 4 | 412.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 1 | 4 | 543.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 1 | 4 | 198.9 | R0.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 1 | 4 | 156.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 1 | 4 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | jhb-cpt-5-1-leg-1 | 412.0 | R609.86 | R2963.94 | R-2354.08 | R100.00 | NO | | courier-2 | jhb-cpt-5-1-leg-2 | 543.2 | R762.01 | R3825.32 | R-3063.31 | R100.00 | NO | | courier-3 | jhb-cpt-5-1-leg-3 | 198.9 | R339.37 | R1372.87 | R-1033.50 | R100.00 | NO | | courier-4 | jhb-cpt-5-1-leg-4 | 156.5 | R290.42 | R1092.83 | R-802.41 | R100.00 | NO | | courier-5 | jhb-cpt-5-1-leg-5 | 110.1 | R238.34 | R796.79 | R-558.45 | R100.00 | NO |

JHB → Cape Town · 5 legs · 5 parcel(s)

- Customer revenue: **R9500.00**; Mzansi gross platform revenue: **R1900.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R1623.00**.
- Total courier pool: **R7600.00**; route cost floor: **R10051.75**; route surplus after courier cost floor: **R-2451.75**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **fails**; dispatch-time gate: **REPRICE**; decision: **REJECT / CONSOLIDATE**.
- Minimum customer tariff at this parcel count: **R2512.94 per parcel**; minimum courier pool: **R10051.75**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R2512.94 per parcel** (courier cost floor R2512.94, positive platform net R1.75).
- Readiness reasons: Courier pool does not cover operating cost plus disruption reserve At least one courier is below the R150.00 net acceptance floor; warnings: Courier pool is R2451.75 below the combined operating-cost floor after deadhead, return, and handover costs.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 5 | 20 | 412.0 | R220.00 | 25 | none | 20 parcels / 250kg | |

Bloemfontein → Beaufort West | 543.2 | 5 | 20 | 543.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 5 | 20 | 198.9 | R0.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 5 | 20 | 156.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 5 | 20 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | jhb-cpt-5-5-leg-1 | 412.0 | R2180.46 | R2963.94 | R-783.48 | R100.00 | NO | | courier-2 | jhb-cpt-5-5-leg-2 | 543.2 | R2801.30 | R3825.32 | R-1024.02 | R100.00 | NO | | courier-3 | jhb-cpt-5-5-leg-3 | 198.9 | R1076.75 | R1372.87 | R-296.12 | R100.00 | NO | | courier-4 | jhb-cpt-5-5-leg-4 | 156.5 | R877.01 | R1092.83 | R-215.82 | R100.00 | NO | | courier-5 | jhb-cpt-5-5-leg-5 | 110.1 | R664.48 | R796.79 | R-132.31 | R100.00 | NO |

JHB → Cape Town · 5 legs · 10 parcel(s)

- Customer revenue: **R19000.00**; Mzansi gross platform revenue: **R3800.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R3247.50**.
- Total courier pool: **R15200.00**; route cost floor: **R10051.75**; route surplus after courier cost floor: **R5148.25**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R1256.47 per parcel**; minimum courier pool: **R10051.75**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R1256.47 per parcel** (courier cost floor R1256.47, positive platform net R0.88).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 10 | 40 | 412.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 10 | 40 | 543.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 10 | 40 | 198.9 | R0.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 10 | 40 | 156.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 10 | 40 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | jhb-cpt-5-10-leg-1 | 412.0 | R4407.44 | R2963.94 | R1443.50 | R100.00 | YES | | courier-2 | jhb-cpt-5-10-leg-2 | 543.2 | R5692.83 | R3825.32 | R1867.51 | R100.00 | YES | | courier-3 | jhb-cpt-5-10-leg-3 | 198.9 | R2122.28 | R1372.87 | R749.41 | R100.00 | YES | | courier-4 | jhb-cpt-5-10-leg-4 | 156.5 | R1708.73 | R1092.83 | R615.90 | R100.00 | YES | | courier-5 | jhb-cpt-5-10-leg-5 | 110.1 | R1268.72 | R796.79 | R471.93 | R100.00 | YES |

JHB → Cape Town · 5 legs · 20 parcel(s)

- Customer revenue: **R31000.00**; Mzansi gross platform revenue: **R6200.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R5299.50**.
- Total courier pool: **R24800.00**; route cost floor: **R10051.75**; route surplus after courier cost floor: **R14748.25**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R628.24 per parcel**; minimum courier pool: **R10051.75**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R628.23 per parcel** (courier cost floor R628.23, positive platform net R0.44).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 20 | 80 | 412.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 20 | 80 | 543.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 20 | 80 | 198.9 | R0.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 20 | 80 | 156.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 20 | 80 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | jhb-cpt-5-20-leg-1 | 412.0 | R7220.46 | R2963.94 | R4256.52 | R100.00 | YES | | courier-2 | jhb-cpt-5-20-leg-2 | 543.2 | R9345.30 | R3825.32 | R5519.98 | R100.00 | YES | | courier-3 | jhb-cpt-5-20-leg-3 | 198.9 | R3442.96 | R1372.87 | R2070.09 | R100.00 | YES | | courier-4 | jhb-cpt-5-20-leg-4

| 156.5 | R2759.33 | R1092.83 | R1666.50 | R100.00 | YES | | courier-5 | jhb-cpt-5-20-leg-5 | 110.1 | R2031.95 | R796.79 | R1235.16 | R100.00 | YES |

JHB → Cape Town · 5 legs · 50 parcel(s)

- Customer revenue: **R65000.00**; Mzansi gross platform revenue: **R13000.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R11113.50**.
- Total courier pool: **R52000.00**; route cost floor: **R10051.75**; route surplus after courier cost floor: **R41948.25**.
- Capacity feasible: **no**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **CONSOLIDATE**; decision: **REJECT / CONSOLIDATE**.
- Parallel planning: **3 wave(s)**; wave decisions: W1 20 parcels / R20800.00 pool / R10748.25 courier net, W2 20 parcels / R20800.00 pool / R10748.25 courier net, W3 10 parcels / R10400.00 pool / R348.25 courier net; planner result: **CONSOLIDATE**.
- Minimum customer tariff at this parcel count: **R251.30 per parcel**; minimum courier pool: **R10051.75**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R251.29 per parcel** (courier cost floor R251.29, positive platform net R0.18).
- Readiness reasons: Vehicle capacity, parcel count, weight, or volume is not feasible; warnings: One or more relay legs exceed the vehicle’s verified parcel, weight, or volume capacity.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 50 | 200 | 412.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 50 | 200 | 543.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 50 | 200 | 198.9 | R0.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 50 | 200 | 156.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 50 | 200 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | jhb-cpt-5-50-leg-1 | 412.0 | R15190.68 | R2963.94 | R12226.74 | R100.00 | YES | | courier-2 | jhb-cpt-5-50-leg-2 | 543.2 | R19693.94 | R3825.32 | R15868.62 | R100.00 | YES | | courier-3 | jhb-cpt-5-50-leg-3 | 198.9 | R7184.87 | R1372.87 | R5812.00 | R100.00 | YES | | courier-4 | jhb-cpt-

5-50-leg-4 | 156.5 | R5736.04 | R1092.83 | R4643.21 | R100.00 | YES | | courier-5 | jhb-cpt-5-50-leg-5 | 110.1 | R4194.47 | R796.79 | R3397.68 | R100.00 | YES |

JHB → Cape Town · 5 legs · 100 parcel(s)

- Customer revenue: **R130000.00**; Mzansi gross platform revenue: **R26000.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R22228.50**.
- Total courier pool: **R104000.00**; route cost floor: **R10051.75**; route surplus after courier cost floor: **R93948.25**.
- Capacity feasible: **no**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **CONSOLIDATE**; decision: **DISPATCH**.
- Parallel planning: **5 wave(s)**; wave decisions: W1 20 parcels / R20800.00 pool / R10748.25 courier net, W2 20 parcels / R20800.00 pool / R10748.25 courier net, W3 20 parcels / R20800.00 pool / R10748.25 courier net, W4 20 parcels / R20800.00 pool / R10748.25 courier net, W5 20 parcels / R20800.00 pool / R10748.25 courier net; planner result: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R125.65 per parcel**; minimum courier pool: **R10051.75**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R125.65 per parcel** (courier cost floor R125.65, positive platform net R0.09).
- Readiness reasons: Vehicle capacity, parcel count, weight, or volume is not feasible; warnings: One or more relay legs exceed the vehicle’s verified parcel, weight, or volume capacity.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 100 | 400 | 412.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 100 | 400 | 543.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 100 | 400 | 198.9 | R0.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 100 | 400 | 156.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 100 | 400 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | jhb-cpt-5-100-leg-1 | 412.0 | R30427.88 | R2963.94 | R27463.94 | R100.00 | YES | | courier-2 | jhb-cpt-5-100-leg-2 | 543.2 | R39478.12 | R3825.32 | R35652.80 | R100.00 | YES | | courier-3 | jhb-

cpt-5-100-leg-3 | 198.9 | R14338.52 | R1372.87 | R12965.65 | R100.00 | YES | | courier-4 | jhb-cpt-5-100-leg-4 | 156.5 | R11426.79 | R1092.83 | R10333.96 | R100.00 | YES | | courier-5 | jhb-cpt-5-100-leg-5 | 110.1 | R8328.69 | R796.79 | R7531.90 | R100.00 | YES |

JHB → Cape Town · 5 legs · 120 parcel(s)

- Customer revenue: **R156000.00**; Mzansi gross platform revenue: **R31200.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R26674.50**.
- Total courier pool: **R124800.00**; route cost floor: **R10051.75**; route surplus after courier cost floor: **R114748.25**.
- Capacity feasible: **no**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **CONSOLIDATE**; decision: **DISPATCH**.
- Parallel planning: **6 wave(s)**; wave decisions: W1 20 parcels / R20800.00 pool / R10748.25 courier net, W2 20 parcels / R20800.00 pool / R10748.25 courier net, W3 20 parcels / R20800.00 pool / R10748.25 courier net, W4 20 parcels / R20800.00 pool / R10748.25 courier net, W5 20 parcels / R20800.00 pool / R10748.25 courier net, W6 20 parcels / R20800.00 pool / R10748.25 courier net; planner result: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R104.71 per parcel**; minimum courier pool: **R10051.75**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R104.71 per parcel** (courier cost floor R104.71, positive platform net R0.07).
- Readiness reasons: Vehicle capacity, parcel count, weight, or volume is not feasible; warnings: One or more relay legs exceed the vehicle’s verified parcel, weight, or volume capacity.

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 120 | 480 | 412.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 120 | 480 | 543.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 120 | 480 | 198.9 | R0.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 120 | 480 | 156.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 120 | 480 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | jhb-cpt-5-120-leg-1 | 412.0 | R36522.75 | R2963.94 | R33558.81 | R100.00 | YES | | courier-2 | jhb-

cpt-5-120-leg-2 | 543.2 | R47391.78 | R3825.32 | R43566.46 | R100.00 | YES | | courier-3 | jhb-cpt-5-120-leg-3 | 198.9 | R17199.98 | R1372.87 | R15827.11 | R100.00 | YES | | courier-4 | jhb-cpt-5-120-leg-4 | 156.5 | R13703.09 | R1092.83 | R12610.26 | R100.00 | YES | | courier-5 | jhb-cpt-5-120-leg-5 | 110.1 | R9982.40 | R796.79 | R9185.61 | R100.00 | YES |

JHB → Cape Town · uneven 4-leg split · 20 parcels

- Customer revenue: **R33000.00**; Mzansi gross platform revenue: **R6600.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R5641.50**.
- Total courier pool: **R26400.00**; route cost floor: **R9990.04**; route surplus after courier cost floor: **R16409.96**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R624.38 per parcel**; minimum courier pool: **R9990.04**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R624.38 per parcel** (courier cost floor R624.38, positive platform net R0.44).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 20 | 80 | 412.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 20 | 80 | 543.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Worcester | 355.1 | 20 | 80 | 355.1 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 20 | 80 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | jhb-cpt-4-20-leg-1 | 412.0 | R7763.44 | R2963.94 | R4799.50 | R100.00 | YES | | courier-2 | jhb-cpt-4-20-leg-2 | 543.2 | R10050.31 | R3825.32 | R6224.99 | R100.00 | YES | | courier-3 | jhb-cpt-4-20-leg-3 | 355.1 | R6406.97 | R2403.99 | R4002.98 | R100.00 | YES | | courier-4 | jhb-cpt-4-20-leg-4 | 110.1 | R2179.28 | R796.79 | R1382.49 | R100.00 | YES |

JHB → Cape Town · 6 legs · 20 parcels

- Customer revenue: **R36000.00**; Mzansi gross platform revenue: **R7200.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R6154.50**.
- Total courier pool: **R28800.00**; route cost floor: **R10761.78**; route surplus after courier cost floor: **R18038.22**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R672.62 per parcel**; minimum courier pool: **R10761.78**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R672.61 per parcel** (courier cost floor R672.61, positive platform net R0.44).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Pretoria | 55.8 | 20 | 80 | 55.8 | R220.00 | 25 | none | 20 parcels / 250kg | | Pretoria → Bloemfontein | 454.6 | 20 | 80 | 454.6 | R180.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 20 | 80 | 543.2 | R0.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 20 | 80 | 198.9 | R0.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 20 | 80 | 156.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 20 | 80 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | jhb-cpt-6-20-leg-1 | 55.8 | R1597.80 | R613.35 | R984.45 | R100.00 | YES | | courier-2 | jhb-cpt-6-20-leg-2 | 454.6 | R8571.37 | R3240.62 | R5330.75 | R100.00 | YES | | courier-3 | jhb-cpt-6-20-leg-3 | 543.2 | R9733.44 | R3645.32 | R6088.12 | R100.00 | YES | | courier-4 | jhb-cpt-6-20-leg-4 | 198.9 | R3722.36 | R1372.87 | R2349.49 | R100.00 | YES | | courier-5 | jhb-cpt-6-20-leg-5 | 156.5 | R2981.60 | R1092.83 | R1888.77 | R100.00 | YES | | courier-6 | jhb-cpt-6-20-leg-6 | 110.1 | R2193.43 | R796.79 | R1396.64 | R100.00 | YES |

JHB → Cape Town · high fuel + empty returns

- Customer revenue: **R33000.00**; Mzansi gross platform revenue: **R6600.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R5641.50**.

- Total courier pool: **R26400.00**; route cost floor: **R19392.91**; route surplus after courier cost floor: **R7007.09**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R1212.06 per parcel**; minimum courier pool: **R19392.91**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R1212.06 per parcel** (courier cost floor R1212.06, positive platform net R0.44).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 20 | 80 | 627.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 20 | 80 | 758.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 20 | 80 | 413.9 | R0.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 20 | 80 | 371.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 20 | 80 | 325.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | stress-fuel-leg-1 | 412.0 | R6788.03 | R4947.20 | R1840.83 | R100.00 | YES | | courier-2 | stress-fuel-leg-2 | 543.2 | R8217.56 | R5926.73 | R2290.83 | R100.00 | YES | | courier-3 | stress-fuel-leg-3 | 198.9 | R4268.52 | R3164.40 | R1104.12 | R100.00 | YES | | courier-4 | stress-fuel-leg-4 | 156.5 | R3808.40 | R2846.18 | R962.22 | R100.00 | YES | | courier-5 | stress-fuel-leg-5 | 110.1 | R3317.49 | R2508.40 | R809.09 | R100.00 | YES |

JHB → Cape Town · strong backhaul

- Customer revenue: **R31000.00**; Mzansi gross platform revenue: **R6200.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R5299.50**.
- Total courier pool: **R24800.00**; route cost floor: **R10801.75**; route surplus after courier cost floor: **R13998.25**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.

- Minimum customer tariff at this parcel count: **R675.11 per parcel**; minimum courier pool: **R10801.75**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R675.11 per parcel** (courier cost floor R675.11, positive platform net R0.44).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 20 | 80 | 462.0 | R220.00 | 25 | R180.00 | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 20 | 80 | 593.2 | R180.00 | 25 | R180.00 | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 20 | 80 | 248.9 | R0.00 | 25 | R180.00 | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 20 | 80 | 206.5 | R0.00 | 25 | R180.00 | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 20 | 80 | 160.1 | R0.00 | 35 | R180.00 | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | stress-backhaul-leg-1 | 412.0 | R7090.79 | R3113.94 | R3976.85 | R100.00 | YES | | courier-2 | stress-backhaul-leg-2 | 543.2 | R9093.74 | R3975.32 | R5118.42 | R100.00 | YES | | courier-3 | stress-backhaul-leg-3 | 198.9 | R3529.98 | R1522.87 | R2007.11 | R100.00 | YES | | courier-4 | stress-backhaul-leg-4 | 156.5 | R2885.57 | R1242.83 | R1642.74 | R100.00 | YES | | courier-5 | stress-backhaul-leg-5 | 110.1 | R2199.92 | R946.79 | R1253.13 | R100.00 | YES |

JHB → Cape Town · one courier cancels / recovery return

- Customer revenue: **R33000.00**; Mzansi gross platform revenue: **R6600.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R5641.50**.
- Total courier pool: **R26400.00**; route cost floor: **R15417.75**; route surplus after courier cost floor: **R10982.25**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **DISPATCH**.
- Minimum customer tariff at this parcel count: **R1059.98 per parcel**; minimum courier pool: **R16959.53**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R963.61 per parcel** (courier cost floor R963.61, positive platform net R0.44).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 20 | 80 | 492.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 20 | 80 | 623.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 20 | 80 | 638.9 | R350.00 | 25 | none | 20 parcels / 250kg | | Laingsburg → Worcester | 156.5 | 20 | 80 | 236.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 20 | 80 | 190.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | stress-cancel-leg-1 | 412.0 | R6160.27 | R3491.94 | R2668.33 | R100.00 | YES | | courier-2 | stress-cancel-leg-2 | 543.2 | R7738.05 | R4353.32 | R3384.73 | R100.00 | YES | | courier-3 | stress-cancel-leg-3 | 198.9 | R7346.40 | R4626.87 | R2719.53 | R100.00 | YES | | courier-4 | stress-cancel-leg-4 | 156.5 | R2847.70 | R1620.83 | R1226.87 | R100.00 | YES | | courier-5 | stress-cancel-leg-5 | 110.1 | R2307.58 | R1324.79 | R982.79 | R100.00 | YES |

JHB → Cape Town · 50% refund / failed delivery

- Customer revenue: **R16500.00**; Mzansi gross platform revenue: **R3300.00**; estimated Mzansi net contribution after 2.9% + R1.50 payment cost: **R2820.00**.
- Total courier pool: **R13200.00**; route cost floor: **R10051.75**; route surplus after courier cost floor: **R3148.25**.
- Capacity feasible: **yes**; minimum payouts funded: **yes**; courier acceptance test: **passes**; dispatch-time gate: **DISPATCH**; decision: **REJECT / CONSOLIDATE**.
- Minimum customer tariff at this parcel count: **R691.06 per parcel**; minimum courier pool: **R11056.92**; maximum viable vehicle capacity: **20 units**.
- Break-even tariff at this parcel count: **R628.23 per parcel** (courier cost floor R628.23, positive platform net R0.44).
- Readiness reasons: none; warnings: none

Relay legs

| Leg | Road km | Parcels | Weight kg | Driven km | Tolls | Handover min | Backhaul | Capacity | | Sandton → Bloemfontein | 412.0 | 20 | 80 | 412.0 | R220.00 | 25 | none | 20 parcels / 250kg | | Bloemfontein → Beaufort West | 543.2 | 20 | 80 | 543.2 | R180.00 | 25 | none | 20 parcels / 250kg | | Beaufort West → Laingsburg | 198.9 | 20 | 80 | 198.9 | R0.00 | 25 | none | 20 parcels /

250kg | | Laingsburg → Worcester | 156.5 | 20 | 80 | 156.5 | R0.00 | 25 | none | 20 parcels / 250kg | | Worcester → Cape Town | 110.1 | 20 | 80 | 110.1 | R0.00 | 35 | none | 20 parcels / 250kg |

Courier settlements

| Courier | Leg | Road km | Gross | Operating cost | Net | Min payout | Accept? | | courier-1 | stress-refund-leg-1 | 412.0 | R3821.39 | R2963.94 | R857.45 | R100.00 | YES | | courier-2 | stress-refund-leg-2 | 543.2 | R4931.90 | R3825.32 | R1106.58 | R100.00 | YES | | courier-3 | stress-refund-leg-3 | 198.9 | R1847.14 | R1372.87 | R474.27 | R100.00 | YES | | courier-4 | stress-refund-leg-4 | 156.5 | R1489.86 | R1092.83 | R397.03 | R100.00 | YES | | courier-5 | stress-refund-leg-5 | 110.1 | R1109.71 | R796.79 | R312.92 | R100.00 | YES |

Features that are working correctly in the hardened model

- The 20% platform allocation and 80% courier pool are conserved once across all relay legs.
- Actual road-route geometry is resolved before the dispatch gate and the same route distances feed economics.
- Verified vehicle capacity, parcel weight/volume, operating costs, reserves, courier net floors, tolls, handovers, deadhead, returns, and confirmed backhaul are included.
- 50+ parcel requests are split into capacity-safe parallel waves; each wave receives its own economic decision.
- The append-only parcel ledger records payment, platform allocation, courier pool, reserves, allocations, refunds, cancellations, replacement costs, payment/payout costs, and final contributions.

Model conclusions

Critical launch findings

1. **Capacity is a hard constraint.** A single courier/leg cannot carry 50–100 parcels when verified capacity is 20; those scenarios correctly reject and must be dispatched as multiple waves or parallel legs.
2. **Long-distance one-parcel routes are commercially fragile.** The 20% platform fee can remain positive while courier net is negative; platform-positive does not mean network-viable.
3. **The 80% pool is conserved once across all couriers.** It is not paid 80% per courier.

4. **Road distance improves allocation accuracy, but route geometry is a dispatch prerequisite.** Missing geometry is rejected rather than estimated.
5. **Backhaul materially improves courier net only when it is real, confirmed work; it must not be treated as guaranteed revenue.**

Required safeguards before launch

- Reject a route unless every leg has current road geometry, a verified vehicle-cost profile, and parcel/weight/volume capacity.
- Reject or consolidate when any courier's estimated net is below zero, below the configured minimum payout, or below a launch target such as R150 per accepted leg.
- Require total courier pool >= max(total operating-cost floor, sum minimum payouts) plus a disruption reserve.
- Treat cancellations, refunds, failed handovers, and replacement legs as append-only ledger events. Never pay or count reserved funds as profit; require an immutable event key for every adjustment.
- Use multi-wave planning for parcel counts above a vehicle's capacity; do not place 100 parcels on one relay leg.
- Reprice or delay dispatch when tolls, deadhead, return distance, fuel price, or handover time moves any leg below its acceptance floor.

Minimum viable operating model

Start with local and regional routes, 5–20 parcels per corridor, verified cars/vans, low deadhead, pre-approved handover points, and confirmed backhaul only. Offer long-distance JHB→Cape Town only at a tariff above the scenario break-even tariff and with 4–6 pre-accepted couriers plus a recovery courier.

Important technical gaps exposed

- Dispatch-time route, capacity, reserve, and courier-net checks are now hard gates; keep route results auditable with the dispatch run.
- Automated external payment-provider refund webhooks still need to call the append-only refund event path before automated payout is enabled.
- Replacement-leg reserve creation is wired to courier-reported failed handovers; the final replacement-leg cost must be recorded before settlement approval.
- Multi-wave planning is implemented as a readiness planner; production corridor execution still needs an operational wave/run record before live 50+ parcel dispatch.